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BSCS - 3B

Set Cartesian Product

If **A** and **B** are two sets, then their cartesian product is defined as - $A \times B := \{(x,y): x \in A, y \in B\}$

Example of Set Cartesian Product

$$A = \{1,2\}$$

$$B = \{a,b\}$$

$$A \times B = \{(1,a),(1,b),(2,a),(2,b)\}$$

Regular Expression

Regular Expressions are straightforward to explain the language that finite automata can understand. It is the best method for illustrating any language. Regular languages are the languages that some regular expressions support. A series of patterns that define a string can also be used to construct a regular expression.

he four (4) rules of Regular Expression:

1. Atom
2. Alternation / Union
3. Concatenation
4. Closure

Examples of different rule of Regular Expression

1. Atom: is a regular expression consisting of only one symbol

Example: R.E String Generated

a a
b b
c c
d d

2. **Alternation**: string is generated by reading any expression, with a symbol of group that is being separated from one another by the alternation symbol.

Example: R.E String Generated

a + b a	b
a + b + c a	b
	c

3. **Concatenation**: string is generated by placing the symbols side by side and on manner the symbols are arranged within the R.E.

Example: R.E String Generated

abc abc
cba cba
abba abba
abbc abbc

4. Closure (Kleene Star & Positive Closure)

Kleene Star: string can be generated by simply reading the symbol or symbols group together and raised with an ‘ * ’.

Note: that the symbols ‘ε’ will be used to designate empty reading or transition.

Example: R.E String Generated

a^* aaaa
aaaaaaa
e
a

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Positive Closure: strings can be generated as like that of the Kleene Star. Except the 'e' reading.

Example: R.E String Generated

a^+ aaaaa
aaaaaaaaaa
a
aa

Example: Combining the rules in creating Regular Expression(R, E), we can therefore create the following Regular Expression (R. E).

1. $ab + a^* ab$

String Generated

aaaa
aaa
a

2. $ab^+ + a^* abbbbbb ab$

abb
aaaaaaa
a
aa

$$3. (a+b)^* + a (b + c^+) e$$

aaa

bbbbbb

ab

accccccc

accc

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Exercise 1: Combining the rules of Regular Expression (R.E), find the string generated base to the given Regular Expression.

Note: Give at least five (5) string generated.

String Generated

$$1. a(a+b)^+ + abc (a^* + bc^*)$$

1. aa

2. ab

3. aab

4. abca

5. abcbc

$$2. ab^+(ab(a+b)^* + b^+(ab)^*)abc^+$$

1. abababc

2. ababbabc

3. abbbabc

4. ababababc

5. Abbbababc

$$3. ab^*c^+(ab + c^*(a^+b^* + (abc)^*))bc$$

1. acabbc

2. acabc

3. acbbc

4. acabcbc

5. abcabc

4. $(ab)^+(ab + ac)^* + ab^* (bc (a+b)^*)$

1. ab
2. abab
3. abac
4. abbc
5. abbca

5. $ab^*(abc(ab^*(b+a)^+ + ab^*(a+b)^+)a^*)$

1. abcabb
2. abcaba
3. abcabba
4. abcabaa
5. abbcabb

6. $11^* (1+0)^* + 111(1+100)^+$

1. 1
2. 11
3. 10
4. 110
5. 1111

7. $1011 (110 + 111)^+$

1. 1011110
2. 1011111
3. 1011110110
4. 1011111111
5. 1011110111

8. $(bc)^+(a + abc)^* + a^* (bc(a+b)^*)$

1. bc
2. bcbc
3. bca
4. bcabc
5. bcabca

9. $(abc^* + a(a+b)^*) + ab^*c^*(a+b)^+$

1. ab
2. abc
3. abcc
4. aa
5. aba

10. $ab^* + b(a+b)^*abc^+$

1. a
2. ab
3. abb
4. babc
5. babcc